# ■■ LUMINA — Technical Stack & System Architecture

**AI-Driven Crime Intelligence & Analytical Platform · Production Architecture Specification**

## ■■ Architectural Tier Breakdown

| Tier | Technology | Specification & Role |
|---|---|---|
| **Frontend UI** | React 19 + TypeScript + Vite 8.1 | High-performance SPA with strict typing, HMR, relative bundling (`base: './'`) |
| **Routing** | @tanstack/react-router | SPA Hash History (`createHashHistory`) ensuring zero 404s on Catalyst `/app/` subpaths |
| **Styling** | Tailwind CSS v4 + Vanilla CSS | Tactical dark mode design system, glassmorphism, responsive data grids |
| **Mapping / GIS** | Leaflet 1.9 + Esri Cartography | Multi-layer raster tiles (Esri Dark Canvas, Tactical Midnight, Satellite) |
| **Network Graphs** | Cytoscape.js | Interactive accused-case link analysis, syndicate hub detection |
| **Visualizations** | ECharts & Custom SVG | Statewide crime distribution, trend forecasting, status donuts |
| **Serverless Compute** | Zoho Catalyst Advanced I/O | Python 3.13 serverless runtime (`api_service`) with sub-second execution |
| **Batch Processing** | Zoho Catalyst Cron | Nightly scheduled data ingestion and database schema validation (`etl_cron`) |
| **Data Storage** | Catalyst Data Store + SQLite | Dual-Engine: ZCQL cloud storage with resilient in-memory local fallback (5,005 FIRs) |
| **AI Backbone** | Google Gemini 2.5 Flash | Bilingual Generative AI Copilot with RAG database context injection |
| **Neural Audio** | Google Neural TTS | Bilingual voice synthesis with 5 calibrated speed presets (0.75x to 1.70x) |

## ■ Core Algorithms & Intelligence Models

### 1. Spatiotemporal Hotspot Detection (ST-DBSCAN)

Lumina computes real-time incident density using the spherical **Haversine Distance formula** coupled with temporal day-delta windows: $d = 2R \cdot \arcsin(\sqrt{\sin^2(\Delta\phi/2) + \cos(\phi■)\cos(\phi■)\sin^2(\Delta\lambda/2)})$. The algorithm groups incidents within $\varepsilon_{spatial}$ (500m–2.5km) and $\varepsilon_{temporal}$ (45 days) to dynamically isolate active crime corridors and assign tactical patrol routes.

### 2. Threat & Risk Scoring Algorithm

Dynamic threat scoring (0–100 scale) calibrated using non-linear incident density and violent crime proportions: Threat Score = min(98, max(28, 18.0 + 58.0·√(ClusterSize/MaxSize) + 26.0·(ViolentRatio))). Suspects with risk scores ≥ 85 are automatically promoted to the supervisory watchlist.

### 3. Bilingual Retrieval-Augmented Generation (RAG)

When an officer queries the copilot, the backend RAG pipeline extracts FIR references, queries the Data Store, and injects factual case records into the Gemini 2.5 Flash system instruction as immutable ground truth. In Kannada mode, official Karnataka Police terminology is enforced without English code-switching.

## ■ Performance Benchmarks & SLA Verification

| Operational Metric | Measured Value | Target Benchmark | Status |
|---|---|---|---|
| **Instant Search Keystroke Latency** | **< 45 ms** | < 100 ms | ■ **Exceeds SLA** |
| **ST-DBSCAN Clustering (5,005 FIRs)** | **< 85 ms** | < 250 ms | ■ **Exceeds SLA** |
| **AI Copilot (RAG + Gemini Inference)** | **850 ms — 1.2 s** | < 2.5 s | ■ **Exceeds SLA** |
| **Official Briefing PDF Compilation** | **220 ms** | < 500 ms | ■ **Exceeds SLA** |
| **Production Gzipped Bundle Size** | **231 KB** | < 500 KB | ■ **Exceeds SLA** |